Selehadin Nasir

Addis Ababa, Ethiopia

+251 965 029742 | +251 704 265868 | nasir.selehadin@gmail.com

selehadin.netlify.app | linkedin.com/in/selehadin-nasir | github.com/SelahNs

Education

Addis Ababa University (AAU) — **Addis Ababa, Ethiopia**

Bachelor of Science in Electrical and Computer Engineering — *Expected Graduation: May 2028*

Projects

CodeDash | *React, Node.js, Express, MongoDB, Socket.io, Redis, Bull, JWT*

- Engineered a custom VS Code extension (VS Code API) that monitors local active files, keystrokes, and session times, utilizing a local offline queue to buffer data during network drops.
- Developed a scalable Node.js/Express backend that receives buffered session payloads and syncs active workspace data over WebSockets (Socket.io) to minimize API overhead.
- Integrated with the GitHub API to fetch and aggregate historical repository data, including commits, programming languages, and user activity metrics.
- Configured a secure GitHub Webhook receiver to handle real-time push and pull-request events, dynamically updating the database with live development logs.
- Implemented background task queues using Bull and Redis to handle asynchronous data synchronization without blocking the main event loop.
- Developed a responsive dark-mode React dashboard utilizing Recharts to visualize user productivity metrics, level progress, and language evolution.

Smart Logic Simulator | *Java, JavaFX, CSS, Object I/O Streams, SVGPath*

- Developed an interactive digital logic simulator in Java and JavaFX, separating UI rendering from underlying logic models.
- Engineered a custom real-time simulation engine using a high-frequency JavaFX AnimationTimer clock to propagate logic states across active gates.
- Designed custom binary serialization (*.ckt) using Object I/O streams, allowing users to save and reload circuit layouts locally.
- Programmed dynamic, vector-rendered connection wires using CubicCurve mathematical modeling to calculate and render wire routing and active logic states (glow effects) on the fly.
- Designed a responsive, dark-mode CSS stylesheet using custom variables, drop shadows, and neon transition properties to deliver a premium user interface experience.
- Rendered scalable, ANSI-standard logic gate vector shapes using inline JavaFX SVGPath strings.

Computer Architecture Design | *Logisim-Evolution*

- Designed and simulated a functional 4-bit Arithmetic Logic Unit (ALU) from scratch to compute logic gates, addition, and subtraction.
- Developed an 8-bit multiplier schematic to perform mathematical operations at the hardware logic level.

Technical Skills

Languages: JavaScript, Java, Python, C (Basic), MATLAB
Web Technologies: React, Node.js, Express, MongoDB, Socket.io, JWT, HTML5/CSS3, Tailwind CSS
Hardware & Tools: Logisim-Evolution, VS Code Extension API, Git, GitHub, Vercel, Netlify, Render, Bull (Redis)